

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 09 · Basic Trigonometry

Questions

Student book · questions with writing space

- 21 pages
- Past papers up to June 2026

Basic Trigonometry

Your classified route

7
questions

42
marks

3
families

5
variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Basic Trigonometry

Families and variants

F01 • Area, angle and side linkage

V01 Area gives $\sin(\theta)$; choose the branch; use cosine rule

V02 Side ratio gives largest angle; area fixes the scale

F02 • Sine rule and the ambiguous case

V01 SSA sine rule with longest-side branch selection

F03 • Multi-step triangle synthesis

V01 Derive a side relation, solve it exactly, then find exact area

V02 Composite triangles: derive cosine expression, then propagate with sine rule and area

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Area, angle and side linkage

WHAT THIS FAMILY ASKS

Move between area, included angle and side length, making the correct acute/obtuse choice before applying the cosine rule.

VARIANT ROUTE

V01 Area gives $\sin(\theta)$; choose the branch; use cosine rule

V02 Side ratio gives largest angle; area fixes the scale

4 Complete Classified questions

21 total marks

4. A parallelogram $ABCD$ has area 40 cm^2

Given that AB has length 10 cm , BC has length 6 cm and angle DAB is obtuse, find

(a) the size of angle DAB , in degrees, to 2 decimal places,

(3)

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4. A parallelogram $ABCD$ has area 40 cm^2

Given that AB has length 10 cm , BC has length 6 cm and angle DAB is obtuse, find

(b) the length of diagonal BD , in cm , to one decimal place.

(2)

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2. The triangle ABC is such that

- $AB = 15 \text{ cm}$
- $AC = 25 \text{ cm}$
- $\text{angle } BAC = \theta^\circ$
- $\text{area triangle } ABC = 100 \text{ cm}^2$

(a) Find the value of $\sin \theta^\circ$

(2)

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2. The triangle ABC is such that

- $AB = 15 \text{ cm}$
- $AC = 25 \text{ cm}$
- angle $BAC = \theta^\circ$
- area triangle $ABC = 100 \text{ cm}^2$

Given that $\theta > 90$

(b) find the length of BC , in cm, to 3 significant figures.

(3)

3. A triangular plot of land, ABC , has area 60 m^2

Given

- $AB = 15 \text{ m}$
- $AC = 10 \text{ m}$
- $\text{angle } BAC = \theta^\circ$

(a) show that $\sin \theta^\circ = \frac{4}{5}$

(1)

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3. A triangular plot of land, ABC , has area 60 m^2

Given

- $AB = 15$ m
- $AC = 10$ m
- angle $BAC = \theta^\circ$

Given also that θ° is the **largest** angle in the triangle,

(b) find the length of the side BC , in metres, to 3 significant figures,

(3)

3. A triangular plot of land, ABC , has area 60 m^2

Given

- $AB = 15 \text{ m}$
- $AC = 10 \text{ m}$
- $\text{angle } BAC = \theta^\circ$

(c) find the size of the **smallest** angle in the triangle, in degrees, to one decimal place.

(2)

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3. The sides of a triangle are in the ratio 2:4:5

(a) Find the size of the **largest** angle of the triangle, in degrees, to one decimal place.

(2)

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3. The sides of a triangle are in the ratio 2:4:5

Given that the area of the triangle is 140 cm^2

(b) find the length of the **shortest** side, in cm, to one decimal place.

(3)

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Sine rule and the ambiguous case

WHAT THIS FAMILY ASKS

Use an SSA configuration to calculate a sine value and decide between supplementary angles from geometric information.

VARIANT ROUTE

V01 SSA sine rule with longest-side branch selection

1 Complete Classified questions

5 total marks

2. In the triangle ABC ,

- $AB = 21$ cm
- $BC = 13$ cm
- angle $BAC = 25^\circ$
- angle $ACB = x^\circ$

(a) Use the sine rule to find the value of $\sin x^\circ$, giving your answer to 4 decimal places. **(2)**

In the triangle ABC ,

- $AB = 21 \text{ cm}$
- $BC = 13 \text{ cm}$
- $\text{angle } BAC = 25^\circ$
- $\text{angle } ACB = x^\circ$

Given also that AB is the longest side of the triangle,

(b) find the value of x , giving your answer to 2 decimal places.

(3)

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Multi-step triangle synthesis

WHAT THIS FAMILY ASKS

Chain cosine rule, algebra, sine rule and area across one or more triangles while preserving exact values as long as possible.

VARIANT ROUTE

V01 Derive a side relation, solve it exactly, then find exact area

V02 Composite triangles: derive cosine expression, then propagate with sine rule and area

2 Complete Classified questions

16 total marks

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

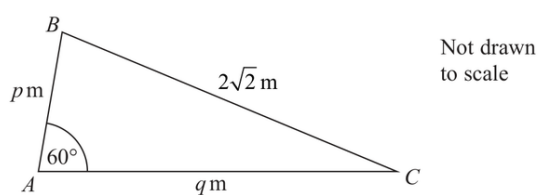


Figure 1

Figure 1 shows the plan view of a flower bed.

The flowerbed is in the shape of a triangle ABC with

- $AB = p$ metres
- $AC = q$ metres
- $BC = 2\sqrt{2}$ metres
- angle $BAC = 60^\circ$

(a) Show that

$$p^2 + q^2 - pq = 8$$

(2)

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

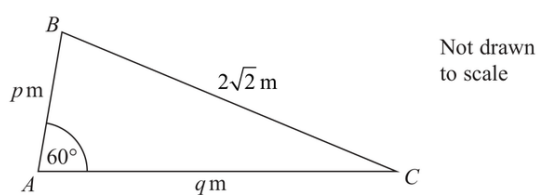


Figure 1

Figure 1 shows the plan view of a flower bed.

The flowerbed is in the shape of a triangle ABC with

- $AB = p$ metres
- $AC = q$ metres
- $BC = 2\sqrt{2}$ metres
- angle $BAC = 60^\circ$

Given that side AC is 2 metres longer than side AB , use algebra to find

- (b) (i) the exact value of p ,
(ii) the exact value of q .

(5)

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

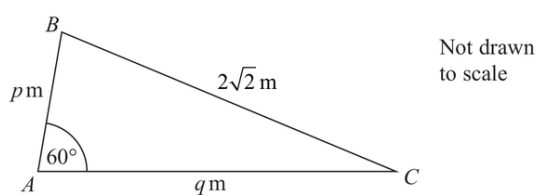


Figure 1

Figure 1 shows the plan view of a flower bed.

The flowerbed is in the shape of a triangle ABC with

- $AB = p$ metres
- $AC = q$ metres
- $BC = 2\sqrt{2}$ metres
- angle $BAC = 60^\circ$

Using the answers to part (b),

(c) calculate the exact area of the flower bed.

(2)

5.

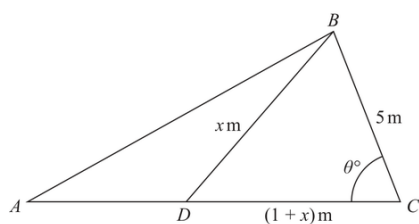


Diagram **NOT**
accurately drawn

Figure 2

Figure 2 shows the plan view of a frame for a flat roof.

The shape of the frame consists of triangle ABD joined to triangle BCD .

Given that

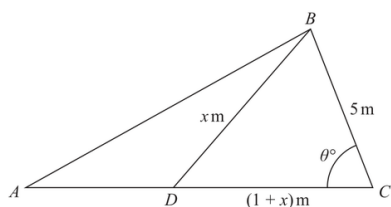
- $BD = x \text{ m}$
- $CD = (1 + x) \text{ m}$
- $BC = 5 \text{ m}$
- angle $BCD = \theta^\circ$

(a) show that $\cos \theta^\circ = \frac{13 + x}{5 + 5x}$

(2)

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Diagram **NOT**
accurately draw



The shape of the frame consists of triangle ABD joined to triangle BCD .

- $BD = x$ m
- $CD = (1 + x)$ m
- $BC = 5$ m
- angle $BCD = \theta^\circ$

- $x = 2\sqrt{3}$
- $\text{angle } BAC = 30^\circ$
- ADC is a straight line

(5)